Verification and Approximation Study for *Generalized Cluster Dynamics* — Campaign Results

RadCluster_2_1

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Abstract

This report collects the results of the verification campaign run in answer to the reviewer's §6.2 / Figure 8 critique. It reports the six experimentally observable parameters for every campaign configuration, against the EUROFER97 measurements where a comparison is meaningful and against the fully discrete solution for the closure verification. Section 4 adds a second campaign that separates system size from binning, two axes the earlier tables varied together. Three defects found *by* the campaigns — a physics channel missing from one solver path, a dose ladder that scored the wrong dose, and a corrupted moment reaching the SIA inventory — are documented alongside the numbers they invalidated, because they bear directly on what the tables may be used to claim.

Contents

1 Campaign Parameters

All runs share the EUROFER97 host declaration, fission cascade spectrum, $G = 1 \times 10^{-7}$ dpa/s, 330 °C, `full_system` solver mode, GMRES with the Woodbury preconditioner, `atol` $= 10^{-20}$, and $i_{\text{mobile}} = v_{\text{mobile}} = 5$. Table 1 lists what differs.

Table 1: Campaign configuration. N_{eq} is the main block; the ⟨100⟩ loop block is carried separately and adds a further I equations in discrete mode (so run D integrates 12 006, not 8 006).

Block	Equations	I	V	i_{discrete}	I_{bin}	Dose (dpa)	Scored at
Table 1 (B1–B4)	bin_moment	80 000	20 000	6400–100	7–18	40	15.72 dpa
Table 2	bin_moment	80 000	20 000	100	10–40	40	15.72 dpa
Table 3	bin_moment	80 000	20 000	100	18	40	15.72 dpa
Table 4 (D)	**discrete**	4 000	4 000	—	—	2	2.000 dpa
Table 4 (C0–C4)	bin_moment	4 000	4 000	1600–5	2–19	2	2.000 dpa

The EUROFER97 neutron measurements at 300 °C to 350 °C (Table 2) are the experimental column throughout. Their dose coverage matters: the loop data begin at 14.6 dpa, which is why Tables 1–3 are scored at 15.72 dpa and why Table 4, which runs to 2 dpa, cannot be compared with them at all.

Table 2: EUROFER97, neutron, 300 °C to 350 °C. Measured ranges over all sources in `docs/Database`.

Population	n	Density (10^{21} m^{-3})	Diameter (nm)
$\frac{1}{2}\langle 111\rangle$ loops	13	0.30 – 34.0	3.3 – 35.0
⟨100⟩ loops	10	2.24 – 12.0	2.8 – 48.0
Cavities	6–8	0.32 – 6.30	1.6 – 16.0
Dose coverage: loops 14.6 dpa to 32 dpa, cavities 2.7 dpa to 32 dpa			

1.1 Table 1 — Closure convergence at the production domain

Table 3: Closure convergence, $I = 80\,000$, $V = 20\,000$, scored at 15.72 dpa. B4 is the tracked reference run. B1 is reported as *d.n.r.* under the rule that a run missing its comparison dose never appears as a metric.

Rung	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
B1 (i_d 6400)	~8030	*d.n.r.* — 0.079 dpa of 40 after 18.9 h						
B2 (i_d 1600)	2048	3.449	4.892	4.156	6.001	2.233	2.232	0.069
B3 (i_d 400)	562	3.512	4.509	4.090	5.868	2.237	2.234	0.068
B4 = production	**189**	**3.498**	**4.637**	**4.715**	**5.939**	**3.609**	**2.890**	**0.060**
Experiment		≥0.30	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16	—

Table 4: One knob changed from B4 at a time, scored at 15.72 dpa.

Column	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
B4 (reference)	189	3.498	4.637	4.715	5.939	3.609	2.890	0.060
$P = 1$ constant	150	2.566	**15.580**	5.335	**45.676**	11.361	2.751	0.158
$P = 3$ lognormal	228	*d.n.r.* — 5.4×10^{-4} dpa of 40; 2.9 h on one output step						
I_{bin} 10	171	3.543	4.282	4.021	5.861	2.900	2.340	0.070
I_{bin} 40	233	*d.n.r.* — 5.70 dpa of 40						
`rtol` 1×10^{-4}	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
`rtol` 1×10^{-6}	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
Experiment		≥0.30	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16	—

1.2 Table 2 — Intra-bin closure and tolerance

1.3 Table 3 — Helium

Table 5: Helium treatment, scored at 15.72 dpa. Model-to-model: the fusion arms have no matching EUROFER97 dataset.

Column	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
fission QSS	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
fission dynamic	190	3.533	4.354	4.153	5.919	3.253	2.434	0.068
fusion QSS	208	2.693	**14.030**	4.155	**51.921**	1.360	0.699	0.158
fusion dynamic	—	*d.n.r.* — 0.052 dpa of 40 after 13.9 h						
fusion Case 1/2	—	blocked on the §3.1 `he_model` unwelding (not implemented)						

1.4 Table 4 — Closure verification against the exact solution

Deviations here are quoted against run D, the fully discrete solution, not against experiment. Both arms run with `LOOP_COAL = 0` so that they solve identical equations; see §5.

Table 6: Closure verification, $I = V = 4\,000$, `LOOP_COAL = 0` in both arms, scored at 2.0000 dpa. Percentages are deviations from run D. N_{eq} is the main block, which is what the solver reports; *total* adds the appended ⟨100⟩ loop block that CVODE also integrates (I equations in discrete mode, $i_{\mathrm{discrete}} + 2I_{\mathrm{bin}}$ in bin-moment mode). The figures in §3 are labelled with the total.

Rung	N_{eq}	total	N_{111} (10^{21})	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)
D (exact)	**8006**	**12006**	**12.568**	**1.065**	**5.741**	**2.948**	**2.544**
C0 (finer than C1)	1226–3214	—	*d.n.r.* — the no-coarsening variant is markedly stiffer the more of the frozen spectrum is resolved discretely; C1 ran in 12 s, $i_d = 600$–1600 did not complete				
C1 (1/10)	830	1242	12.335 (−1.9%)	1.044 (−2.0%)	5.690 (−0.9%)	2.555 (−13.3%)	2.273 (−10.6%)
C2 (1/40)	244	364	12.144 (−3.4%)	1.044 (−2.0%)	5.681 (−1.0%)	2.574 (−12.7%)	2.279 (−10.4%)
C3 (1/160)	108	161	7.027 (−44.1%)	1.018 (−4.4%)	5.349 (−6.8%)	3.219 (+9.2%)	2.451 (−3.6%)
C4 (1/800)	86	129	6.142 (−51.1%)	0.953 (−10.5%)	5.678 (−1.1%)	6.575 (+123.0%)	2.660 (+4.6%)
C4, $P = 1$	51	94	17.498 (+39.2%)	0.059 (−94.5%)	10.755 (+87.3%)	20.929 (+610%)	2.632 (+3.5%)

1.5 Table 4b — The extended rungs against experiment

C2–C4 were rerun to the production horizon so that their curves reach the dose where the EUROFER97 measurements actually sit. At 15.72 dpa they can therefore be compared with experiment directly — something Table 4 cannot do, since it is scored at 2 dpa where there are no loop data at all. Run D has no counterpart here: the discrete arm could not be carried past 0.014 dpa, so **beyond 2 dpa these rungs have no exact reference.** They show dose dependence and the approach to the measured band, not verified accuracy.

Table 7: The no-coarsening rungs carried to 40 dpa, scored at 15.72 dpa against the EUROFER97 measured ranges. *Not* a verification: there is no exact solution at this dose.

Rung	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)
C2 (1/40)	244	13.092	1.722	6.916	5.045	1.951	2.135
C3 (1/160)	108	6.770	1.487	6.392	4.770	2.235	2.230
C4 (1/800)	86	5.276	1.476	5.950	4.748	3.431	2.484
Experiment	—	0.30–34.0	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16

All three land inside the measured $\langle 100\rangle$ density and diameter ranges and inside the cavity diameter range. N_{111} sits above the measured band and d_{111} below it — the signature of the disabled coarsening channel, which is what pins the $\frac{1}{2}\langle 111\rangle$ mean size at the mobility cutoff and inflates its density. That is the expected consequence of running the no-coarsening variant, not an independent finding.

1.6 Table 5 — Null knobs

Table 5 (preconditioner and thread-count null tests) was not run. Its columns measure wall-clock as the dependent variable, so they require an otherwise idle machine, and the campaign machine was occupied throughout. The evidence already in the repository stands in its place: from `compare_linsol_results.json`, the one pair where both arms ran to completion (bin_moment, $N_{eq} = 190$, 0.01 dpa) agrees to **nine significant figures** on $\langle n_i\rangle$ (24.563 925 879 7 Woodbury vs 24.563 925 898 6 Jacobi), seven on $\langle m_v\rangle$ and eight on swelling, with only the wall-clock differing (1.29×).

2 Analysis of Results

Table 1 — the closure has converged in its own parameter

Across B2, B3 and B4 the observables move by a few percent while N_{eq} falls by a factor of ten: d_{100} spans 5.868–6.001 nm and N_{111} 3.449–3.512. That is what convergence in the closure's refinement parameter looks like, and it is the claim Table 1 exists to support. Two qualifications are needed. First, N_{cav} and d_{cav} at B4 (3.609, 2.890) sit above B2/B3 (2.233, 2.232) by more than the loop observables move, so the cavity side is the least converged. Second, the most-resolved rung B1 never ran, so the ladder is anchored at B2 rather than at the intended top; convergence is demonstrated over a 10× span in N_{eq}, not the 40× planned.

Table 2 — tolerance is a null, closure order is not

`rtol` 1×10^{-4} and 1×10^{-6} agree to every digit reported. Two orders of magnitude in integration tolerance move no observable, which establishes that the tolerance is not a free parameter hiding in the results — one of the specific things the reviewer asked about. I_{bin} 10 likewise sits within a few percent of B4. Against that flat background, $P = 1$ (piecewise-constant closure) gives $d_{100} = 45.7$ nm against 5.9 and $d_{111} = 15.6$ against 4.6 — roughly an eightfold error — and its δ_{FP} degrades to 0.158. The existing §6.2 claim about the constant closure is therefore supported, and by a wide margin. $P = 3$ (lognormal) did not run: it reached 5.4×10^{-4} dpa of 40 with a single output step consuming 2.9 h. That is itself the result the plan anticipated, and it is reported as a failure to reach dose rather than as a number.

Table 3 — the QSS reduction is safe at fission, and the fusion column proves it

Fission QSS and fission dynamic agree to every digit: 3.533, 4.354, 4.153, 5.919, 3.253, 2.434 in both. Taken alone that flatness is uninformative — it is equally consistent with a helium treatment that does not matter and with a comparison that was never sensitive. The fusion column is what makes it a result: at the same grid, fusion QSS gives $d_{100} = 51.9$ nm and $d_{111} = 14.0$ nm, an order of magnitude away. The observables *are* sensitive to the helium environment; they are simply insensitive to how helium is time-integrated when, under fission, there is almost none to integrate ($C_{\mathrm{He,tot}} = 1.18 \times 10^{13}\,\mathrm{m}^{-3}$ in the reference run). The fourth column, fusion + dynamic, did not reach dose, so the fusion pair is represented by QSS alone.

Table 4 — the closure reproduces the exact solution, except on cavity density

Against the fully discrete solution, the linear closure holds d_{100} to within 1.1% at every rung from 830 down to 84 equations — a 100× reduction in cost for a one-percent error on that observable. N_{100} is held to 10.5% at the coarsest rung and 2% at the finest.

The exceptions are informative. N_{111} degrades sharply below C2 (−44% at C3, −51% at C4): the $\frac{1}{2}\langle 111\rangle$ population is peaked at small sizes, and once i_{discrete} drops to 25 and then 5 that peak falls inside the binned region where the closure must represent it rather than resolve it. N_{cav} is worse still at C4 (+123%). So the coarsest rung — which is the production resolution — buys its speed at real cost in the two *density* observables, while the *sizes* remain accurate.

Convergence is not monotonic. C1 and C2 are nearly identical, C3 is worse on N_{111} and d_{100}, and C4 recovers on d_{100} while degrading further on N_{cav}. A refinement ladder that does not improve monotonically is not evidence of a converged closure, and this should be understood rather than presented as one. The most likely explanation is that i_{discrete} and the bin partition interact — the rungs hold the bin ratio r fixed but the *location* of the discrete/binned boundary moves relative to the peak of the distribution — but that has not been tested here.

$P = 1$ fails against the exact solution as decisively as it does against B4: +87% on d_{100}, −94.5% on N_{100}.

3 Effects of System Size

Figures 1–5 show the dose dependence of each observable across the ladder, ordered D, C4, C3, C2, C1, C0. d_{111} is excluded. The EUROFER97 measurements and the band spanning them are overlaid.

The curves stop at two different doses, deliberately. C2, C3 and C4 were rerun to 40 dpa so they reach the dose where the measurements sit (14.6 dpa to 32 dpa for loops). D and C1 stop at 2 dpa — the discrete arm could not be carried further at any affordable cost, and C1 was left on its original horizon.

The consequence must be read carefully: **below 2 dpa the coarse rungs can be judged against the exact solution; above it they cannot.** Where the extended curves pass through the experimental band there is no exact reference in the figure, so agreement there speaks to the model, not to the closure. The band is drawn where the data actually are so that the region with neither exact reference nor measurement cannot be mistaken for a validated one.

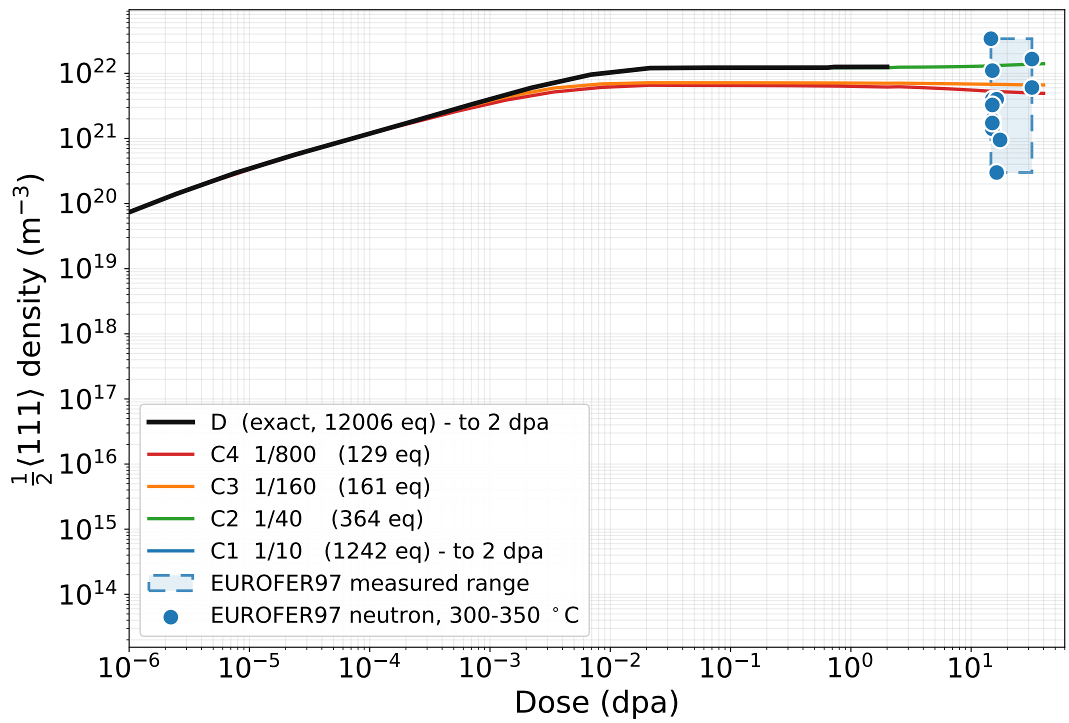

Figure 1: $\frac{1}{2}\langle 111\rangle$ loop number density. The exact solution separates from the coarse rungs from 0.1 dpa onward; C3 and C4 lose roughly half the population.

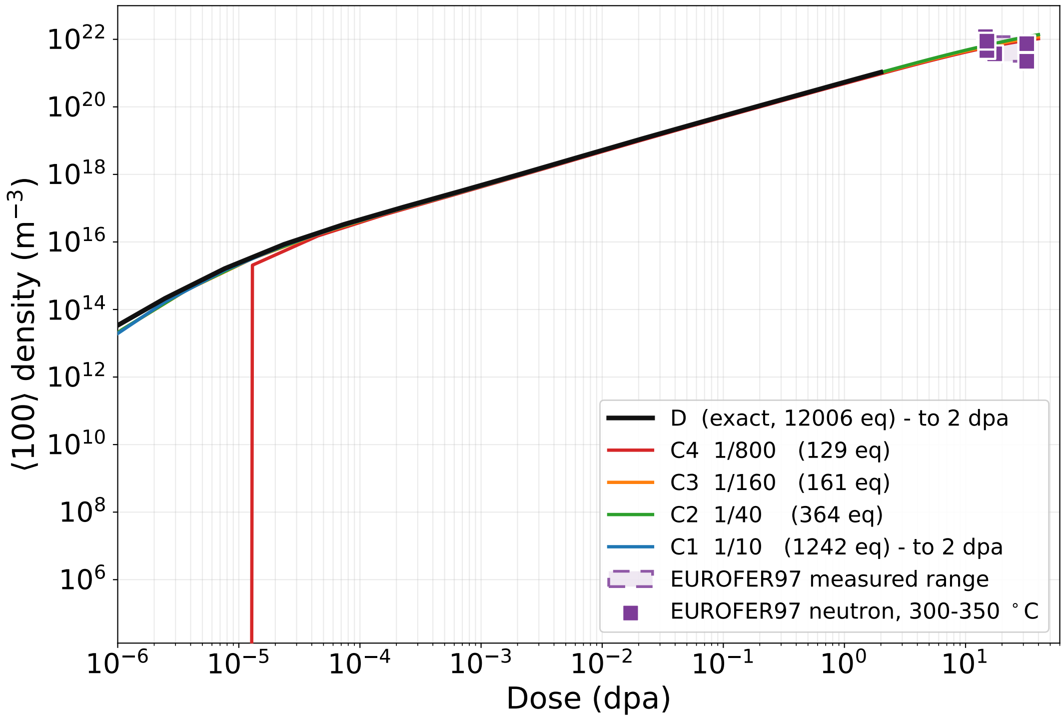

Figure 2: ⟨100⟩ loop number density. All rungs track the exact solution closely; this is the observable the closure represents best.

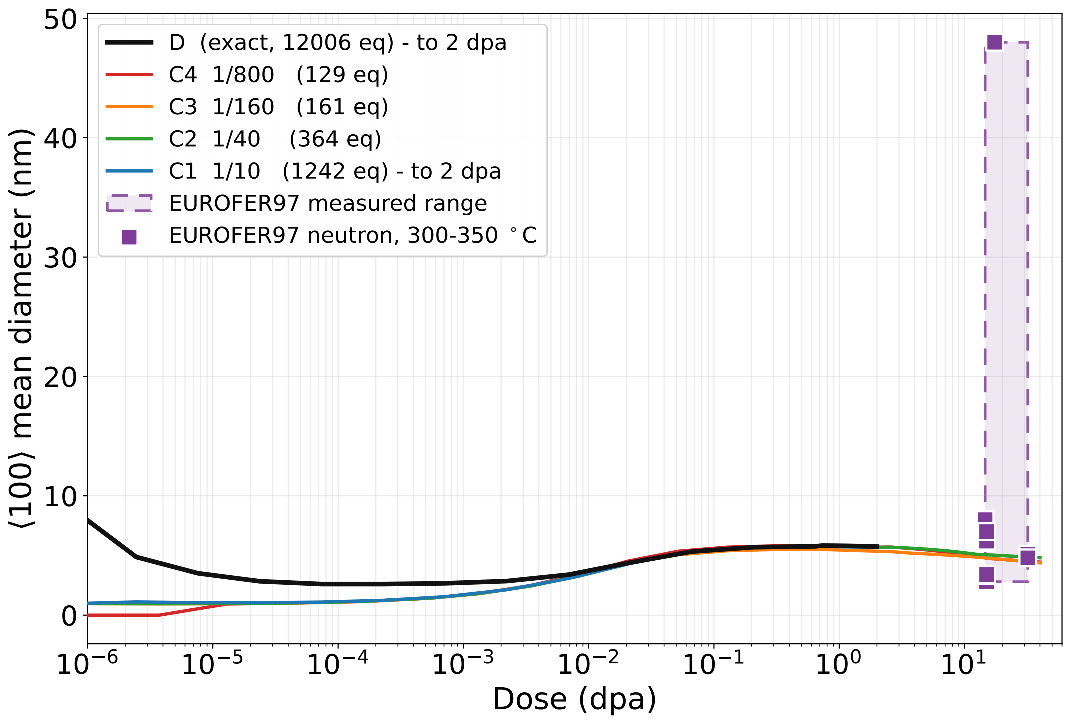

Figure 3: ⟨100⟩ mean diameter. Held to about one percent across the whole ladder.

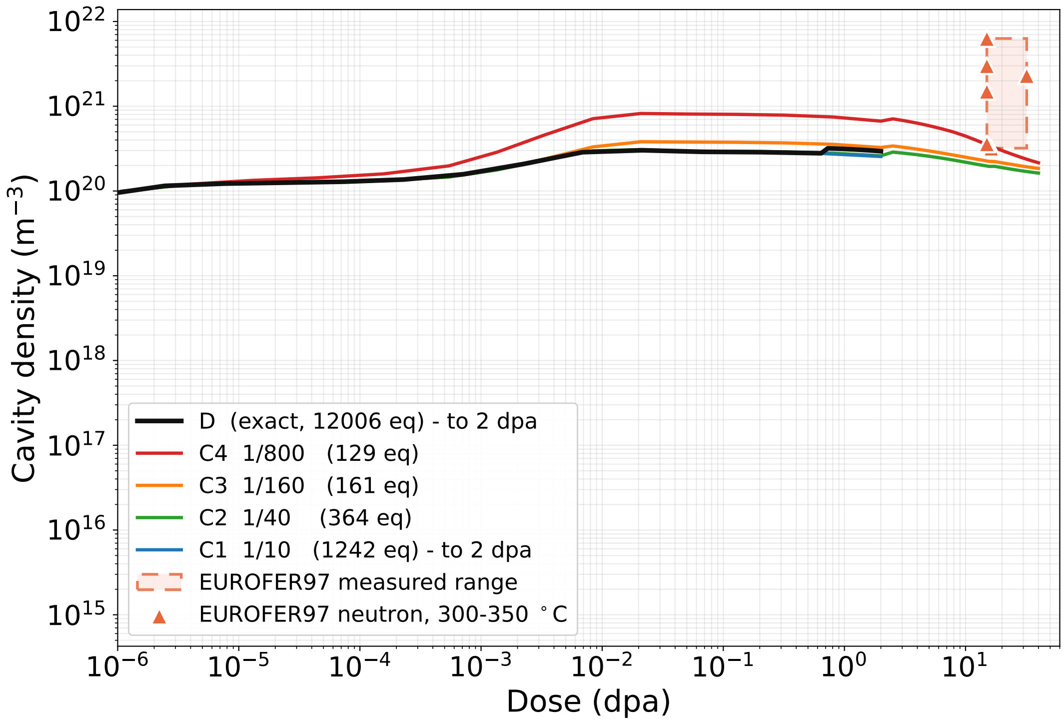

Figure 4: Cavity number density — the observable most sensitive to system size, with C4 more than a factor of two above the exact solution.

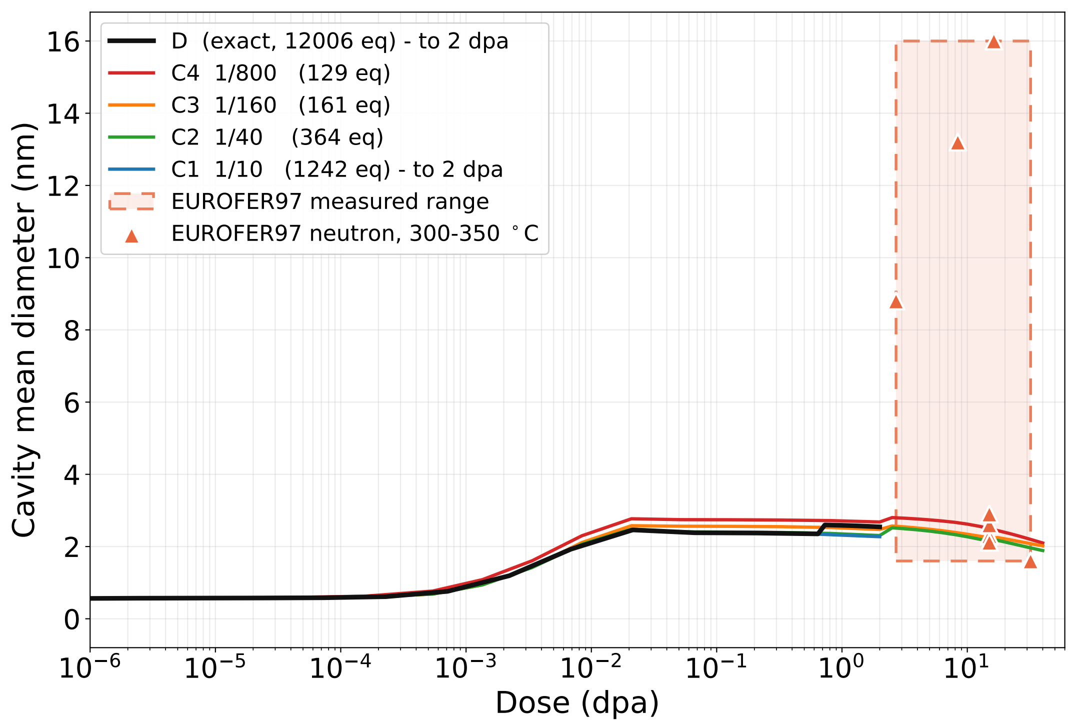

Figure 5: Cavity mean diameter. The cavity data extend down to 2.7 dpa, so this is the one panel where band and curves nearly meet.

4 Campaign M — Separating System Size from Binning

Table 4 measured closure error at one domain and one dose, because that was all its exact arm could reach, and its rungs moved i_{discrete} and I_{bin} *together*. A deviation there cannot be attributed to either. Campaign M separates them: **M1** varies the domain with no closure at all, **M2R** varies the bin count at fixed discrete core, and **M2D** sweeps both independently so that every cell has a neighbour differing in exactly one axis.

All of campaign M runs at production mobility ($i_{\text{mobile}} = v_{\text{mobile}} = 5$), fission cascade, `LOOP_COAL` and `prec_bw` at their defaults in *both* arms, on one shared 45-point logarithmic grid to 20 dpa.

4.1 M1 — what an exact arm costs

Table 8: Cost of the fully discrete arm against domain. The bold row is the only one that completes, and it is the reference for everything below.

Run	$I = V$	N_{eq}	Dose reached	Wall
M1_D1000	1000	2006	20 / 20	5.2 min
M1_D2000	2000	4006	0.03839 / 20	61.1 min
M1_D4000	4000	—	*not run — see text*	

The cliff is abrupt. Doubling the domain takes the exact arm from finishing in five minutes to reaching 0.2 % of the same dose in an hour, and at $I = V = 10\,000$ it diverges outright — per-step cost growing $\approx 4\times$ per output step, 2643 s on step 28 of 45, at 104 % CPU, i.e. entirely inside the serial region. That region is `loop_coal`'s $O(N_C^2)$ pair sum: the participation gate admits sessile sizes above $2\,C_{\text{floor}}$, and at $I = 10\,000$ that population is $\sim 10\times$ the $I = 1000$ one, so the pair count is $\sim 100\times$. It is the same failure that defeated the Table 4 exact arm three times.

Domains of 4000 and above were not attempted. Once 2000 misses by that margin they cannot succeed, and each would cost an hour to confirm it.

4.2 M2R — bin count at the reference domain

Table 9: Deviation from the exact arm (%) at 20 dpa, $I = V = 1000$, $i_{\text{discrete}} = v_{\text{discrete}} = 50$ throughout. Only the bin count varies.

Rung	I_{bin}	r	N_{eq}	N_{111}	d_{111}	N_{100}	d_{100}	d_{cav}
B2	2	4.47	114	+3.7	-4.4	+1.2	-0.6	-2.7
B3	3	2.71	118	+1.8	-5.4	-2.1	+1.0	-2.1
B4	4	2.11	122	+1.4	-5.3	-1.2	-5.4	-2.4
B6	6	1.65	130	+1.2	-5.8	-2.1	-2.3	-1.6
B8	8	1.45	138	+1.1	-6.3	-1.2	-0.8	-0.8
B12	13	1.28	158	+1.0	-4.8	-0.2	+0.1	-0.1
B16	17	1.21	174	+0.9	-5.4	-0.1	+0.3	+0.1
B24	25	1.13	206	+1.0	-6.4	-0.2	+0.1	+0.1
B32	33	1.10	238	+1.0	-6.6	-0.2	+0.1	+0.1

Five of the six observables converge: by $I_{\mathrm{bin}} = 12$ ($N_{\mathrm{eq}} = 158$ against the exact 2006, a 13× reduction) N_{100}, d_{100}, N_{cav} and d_{cav} are all inside 0.3 %, and N_{111} plateaus at a residual +1.0 %.

d_{111} does not. It sits at −5 to −6.6 % and does *not* improve — the finest rung is the worst of the nine. Section 4.4 explains why, and it is not a defect in the closure.

4.3 M2D — the two-dimensional sweep

Table 10: d_{111} deviation from the exact arm (%) at 20 dpa. Read *down* a column, not across a row: the error falls with i_{discrete} and is nearly flat in I_{bin}.

i_{discrete}	$I_{\mathrm{bin}} = 4$	$I_{\mathrm{bin}} = 8$	$I_{\mathrm{bin}} = 16$	$I_{\mathrm{bin}} = 32$
25	-7.2	*d.n.r.*	-6.7	-6.3
50	-5.3	-6.3	-5.4	-6.6
100	*d.n.r.*	-3.4	-4.2	-5.1
200	-6.1	-2.0	-2.5	-3.4
400	*d.n.r.*	-0.1	*d.n.r.*	-1.1
800	+1.0	+0.7	+0.6	*n/r*

Table 11: d_{cav} deviation (%) over the same sweep, for contrast — converged almost everywhere, with one anomaly discussed below.

i_{discrete}	$I_{\mathrm{bin}} = 4$	$I_{\mathrm{bin}} = 8$	$I_{\mathrm{bin}} = 16$	$I_{\mathrm{bin}} = 32$
25	-2.7	*d.n.r.*	+0.1	+0.0
50	-2.4	-0.8	+0.1	+0.1
100	*d.n.r.*	-0.3	-0.0	+0.0
200	+15.1	-0.2	-0.1	-0.1
400	*d.n.r.*	-0.2	*d.n.r.*	-0.1
800	-0.1	-0.1	-0.0	*n/r*

Reading the grid. *d.n.r.* marks a cell that hit the one-hour cap short of 20 dpa and is therefore refused as a metric (§3.4.1); it is a cost result, not a method failure. *n/r* marks a configuration the bin layout cannot realise: at $i_{\mathrm{discrete}} = 800$ only 200 sizes remain for the bins, so $I_{\mathrm{bin}} = 32$ returns 24 and the layout check *raises* rather than running a different grid than the one requested. That guard exists because nine bit-identical "refinement" runs were once produced by a silently dropped bin configuration.

Read *down* column $I_{\mathrm{bin}} = 8$: −6.3, −3.4, −2.0, −0.1, +0.7. Read *across* row $i_{\mathrm{discrete}} = 100$: −3.4, −4.2, −5.1 — worse with every bin added. The bottom row, $i_{\mathrm{discrete}} = 800$, is converged to under 1 % on d_{111} at *every* bin count. The axis that matters is not the one the M2R ladder swept.

One anomalous cell

$i_{\mathrm{discrete}} = 200$, $I_{\mathrm{bin}} = 4$ gives d_{cav} +15.1 % where its row-neighbours sit at −0.2 %, with N_{cav} and $\overline{n}_v$ both +52 %. It ran in 6 s against 39 min to 46 min for those neighbours, and it reproduces exactly on a standalone re-run. Coarse bins make the system non-stiff, which explains the speed; they do not explain the error, because the *coarser* $I_{\mathrm{bin}} = 4$ cells at $i_{\mathrm{discrete}} = 25$ and 50 give −2.7 and −2.4 %. The cause is **not established**. It is recorded here rather than smoothed over, and it is the same non-monotonicity Table 4 showed when C3 came out worse than C2 — with a sharper example.

4.4 Why d_{111} does not converge in bin count

The M2R ladder turned the knob that does not control this error. The $\frac{1}{2}\langle 111\rangle$ mean-size error converges in i_{discrete}, not in I_{bin}, and $i_{\mathrm{discrete}} = 50$ was held fixed across all nine rungs.

At fixed $I_{\mathrm{bin}} = 16$ and 0.05 dpa, against an exact $d_{111} = 2.9787$ nm, the error tracks the fraction of the $\frac{1}{2}\langle 111\rangle$ *content* that lies above the discrete core — i.e. the fraction the closure has to represent:

i_{discrete}		d_{111} (nm)	% of content binned
25	2.7232	(−8.6%)	96.6
50	2.7744	(−6.9%)	91.2
100	2.8128	(−5.6%)	83.4
200	2.8692	(−3.7%)	70.6
400	2.9403	(−1.3%)	47.7
800	2.9813	(+0.1%)	8.0

This is ordinary closure error; it simply converges along the axis that was held fixed. Conversely, at fixed $i_{\mathrm{discrete}} = 400$, raising I_{bin} from 4 to 8 to 33 gives 2.9615, 2.9503, 2.9245 — *slightly worse.*

Nothing is special about $\frac{1}{2}\langle 111\rangle$. Its mean size is ≈ 147 SIAs, about 3× the discrete core, so the peak of the distribution sits inside the binned region and the closure carries it. d_{100} and d_{cav} improve monotonically along the same sweep with about a tenth the amplitude.

Four explanations were tested and rejected, each by measurement:

- **Not a code-path difference.** With $I_{\mathrm{bin}} = 0$ and $i_{\mathrm{discrete}} = I$ — the bin-moment path carrying no closure — it reproduces the discrete arm at 20 dpa to within rounding on all six observables (d_{111} −0.000%, δ_{FP} +0.001%). `check_bm_vs_discrete.py` asks this at 1×10^{-2} dpa; this extends it three decades, to the dose where the deviation is measured.

- **Not `LOOP_COAL`.** It was the leading suspect, being the only channel implemented twice and the only one that moves content up in size space by large jumps. With it *off in both arms* the deficit is unchanged: −7.3% and −7.0%, against −7.0% and −7.9% with it on.

- **Not a domain-edge artefact.** The exact spectrum decays smoothly to the top of the grid — size 1000 holds 0.01 % of the content, with no pile-up — so the edge-piling convention is not implicated.

- **Not slow accumulation.** The deficit is already −8.4% at 0.005 dpa, the first meaningful output point, and drifts only to −6.6% by 20 dpa.

4.5 What this means for choosing a grid

1. **A binning study must sweep both axes.** Table 4 moved them together and confounded them; M2R moved one and hid an error in the other. Neither design can attribute a deviation.

2. i_{discrete} **should be chosen against the population's mean size**, not set to a fixed number. $i_{\mathrm{discrete}} = 50$ with a mean of 147 puts the peak of the distribution inside the closure.

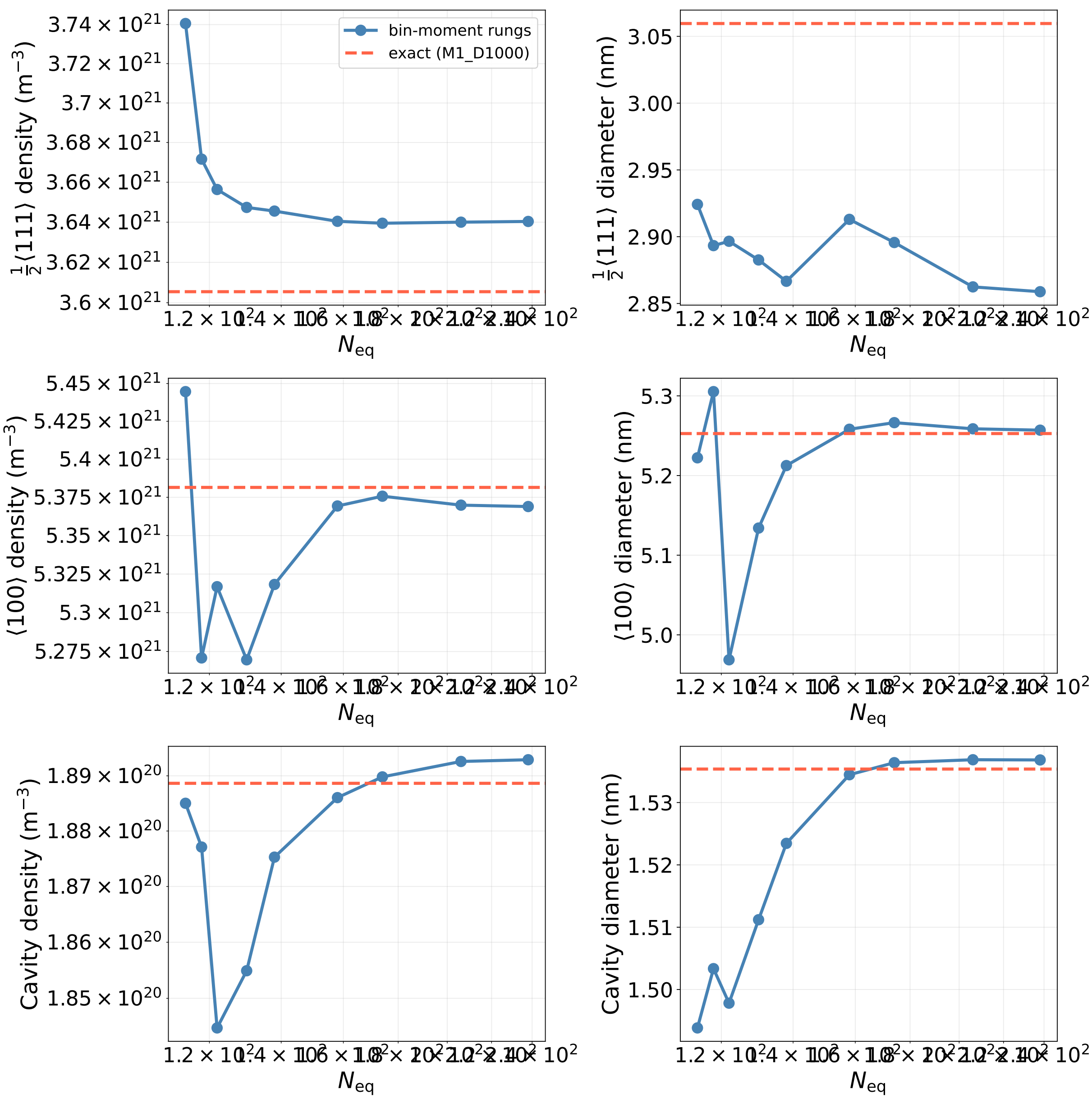

Figure 6: M2R: the six observables against N_{eq} at fixed $i_{\mathrm{discrete}} = 50$, with the exact arm as the dashed line. Five panels close on it; d_{111} runs parallel to it and never approaches.

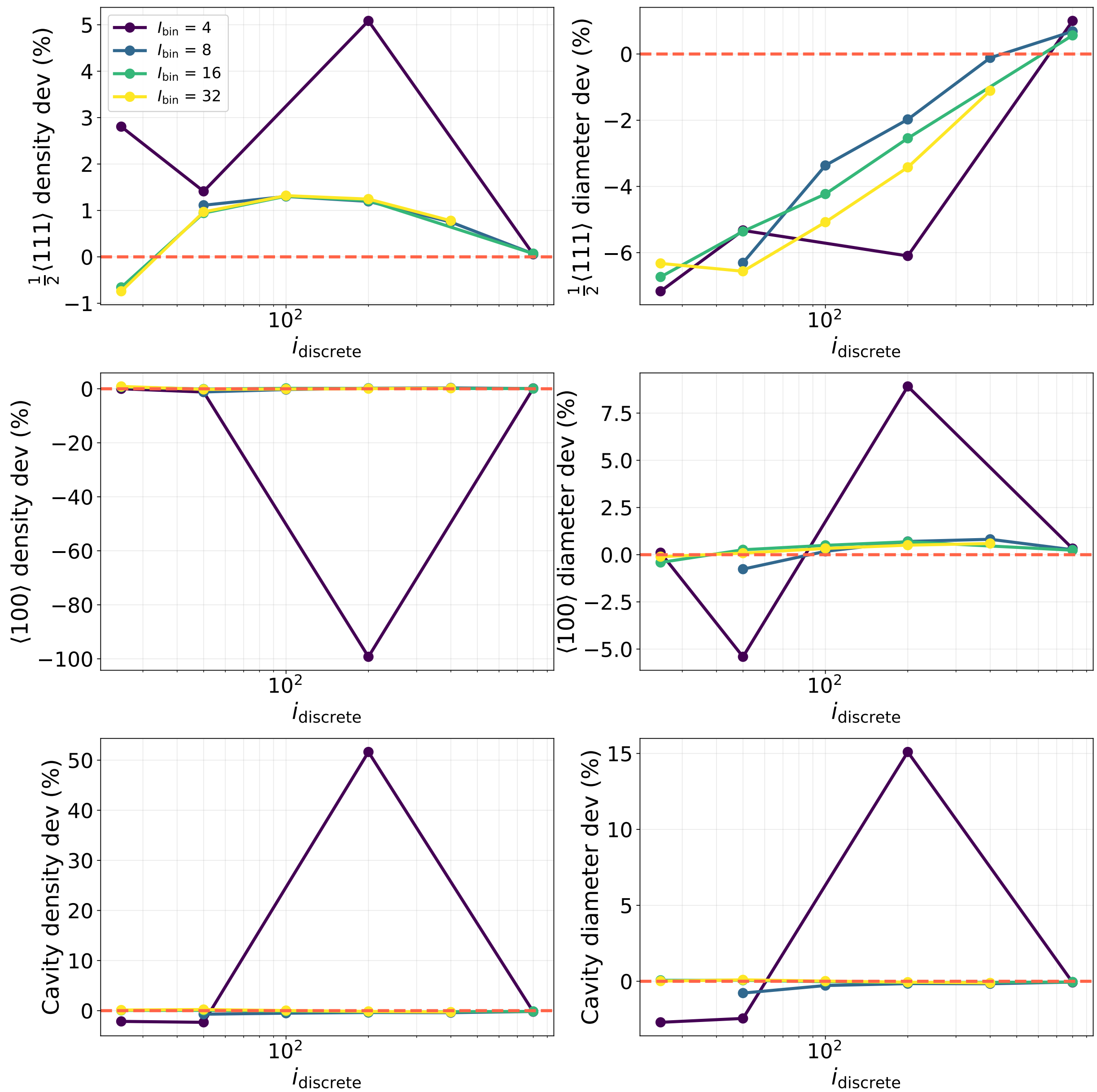

Figure 7: M2D: deviation against i_{discrete}, one curve per I_{bin}. The curves collapse onto one another — the spread *between* bin counts at fixed i_{discrete} is small next to the fall *along* i_{discrete}. This is the figure the M2R ladder could not draw, because it moved along a contour of this surface rather than down it.

3. **Cost is not monotonic in either axis.** At 0.05 dpa the pure discrete solve ran in 13.8 s and $i_{\text{discrete}} = 50$, $I_{\text{bin}} = 16$ in 4.2 s, but $i_{\text{discrete}} = 400$, $I_{\text{bin}} = 16$ took 163.7 s — an order of magnitude slower than either pure form, because the bin-moment path carries an $O(i^2_{\text{discrete}})$ discrete–discrete coalescence block *on top of* the binned machinery. The accurate hybrid is the expensive one.

5 Limitations

1. **Table 4 verifies a variant of the model, not the production configuration.** Production runs with `LOOP_COAL = 1`, the $\frac{1}{2}\langle 111\rangle$ loop–loop coarsening channel. That channel was implemented only in the bin-moment solver path; the discrete path accepted the flag, recorded it in provenance, and ignored it. The discrete arm was therefore never the exact solution of the same equations, and the first version of this table — which showed a 90–114% "closure error" on d_{111} — was measuring the missing channel. The channel has since been implemented in the discrete path and the two paths verified to agree to 7×10^{-8} (`codes/Python_Testing/check_bm_vs_discrete.py`), but the production-configuration exact arm proved unaffordable: the coarsening term is $O(N_C^2)$ in the number of populated sessile sizes and, even gated, spent 2.8 h on a single output step at 0.014 dpa of 2. Table 4 is therefore run with the channel disabled in *both* arms. The closure error it measures is not demonstrably the production run's closure error.

2. **The no-coarsening variant is unphysical.** Without the channel the $\frac{1}{2}\langle 111\rangle$ mean size pins at the mobility cutoff and the loop density runs 20–60× over the EUROFER97 data. This does not invalidate the verification — verification asks whether the closure reproduces the exact solution of the *same* equations, which these two arms do — but nothing in Table 4 may be compared with experiment.

3. **Four columns never reached their comparison dose**: Table 1's B1, Table 2's $P = 3$ and I_{bin} 40, and Table 3's fusion+dynamic. Each is reported with the dose it achieved. The reporting tool refuses to print a metric for such a run, so this rule is mechanical rather than a matter of discipline.

4. $\delta_{\mathrm{FP}} \approx 0.05$–$0.07$ **throughout**, against the model's own 10^{-2} gate, and 0.158 for the two $P = 1$/fusion columns. This is a separate, disclosed defect. A verification study measures closure error against the same equations; conservation error is not that, and the two should not be conflated.

5. $C_{\mathrm{SIA,tot}}$ **is inflated 17–25× in fine-bin runs** ($I_{\mathrm{bin}} \geq 16$). When the top bin empties, its μ_0 falls to the C_{floor} level while its μ_1 does not, leaving $\mu_1/\mu_0 \sim 10^8$ in a bin spanning 946–1001 — a mean size arithmetically impossible for that bin. `post_process` already guards the *mean-size* path (`_floor_bin_moments` clamps μ_1/μ_0 back inside the bin, a guard added after `mean_n_100` once read 2853 on a grid of 1000), but `SIA_content_from_mu1` sums the *raw* μ_1, so the corruption reaches $C_{\mathrm{SIA,tot}}$ unguarded. The six observables, δ_{FP} and swelling are unaffected; what breaks is the swelling identity $S = S_I + \Delta J^d$, since $C_{\mathrm{SIA,tot}}$ is S_I. Reported, not fixed.

6. **Campaign M's reference is a small domain.** $V = 1000$ caps a cavity at about 2.8 nm and the exact arm reports $d_{\mathrm{cav}} = 1.74$ nm at 20 dpa — 62% of the ceiling. Both arms sit on the same truncated domain, so this cancels out of the closure error being measured, but nothing in Section 4 may be compared with experiment.

7. **An earlier dose-ladder defect invalidated a first round of results.** The output grid held no point at the scoring dose, so a requested 15.72 dpa was silently scored at 11.51 dpa (26.8% low) while the run reported having reached it. All affected rows were recomputed. This is the same failure mode as the Figure 8 defect the study was commissioned to answer, found inside the study's own tooling.